

SPAD Measurements

Eduardo Barroso¹ Eikin Kisilkan¹ Utku Karaca²

¹ *École Polytechnique Fédérale de Lausanne (EPFL)*

² *Novoviz*

Abstract

This report documents the experimental setup for the characterization of Single Photon Avalanche Diodes (SPADs). The objective is to classify different light sources by performing coincidence detection experiments. Measurements are conducted using distinct light sources such as pulsed lasers, continuous lasers, and pseudo-thermal light source. Throughout the document, the hardware, optical setups, and results are presented to the reader.

Introduction

Quantum optics classifies light sources based on the photon emission statistics they exhibit. Phase-coherent sources, like lasers, emit independent photons with low intensity fluctuations. The behaviour of this stream of coherent photons is accounted by **Poissonian statistics**. In contrast, thermal sources have larger intensity fluctuations due to the random atomic processes that generate the stream. Thus, this regime is modeled with **super-Poissonian statistics**. The third category is reserved for quantum, single-photon sources. In this regime, photons arrive individually, leading to an effect known as photon anti-bunching. Intensity fluctuations of quantum sources are captured by **sub-poissonian statistics**.

The focus of this work is to measure the statistical properties of these distinct light sources. This is achieved using state-of-the-art single-photon avalanche diodes (SPADs) and time taggers, which enable the recording of individual photon arrivals and the measurement of coincidences between two SPAD channels. To accurately extract the pure statistics of the light sources, the hardware is first rigorously baselined. Specifically, the four primary performance indicators of the SPADs —jitter, photon detection probability (PDP), dark count rate (DCR), and dead time— are quantified us-

ing Time-Correlated Single Photon Counting (TCSPC) techniques [Charbon, 2014, Antolovic et al., 2018, Hofbauer et al., 2018]. With the sensor fully characterized, the setup is then utilized to perform high-resolution coincidence measurements across the different light sources, allowing us to accurately report their intrinsic statistical properties while mathematically accounting for detector artifacts.

0.1 SPAD Performance Indicators

Inside SPADs, a photon incident on the active area of a SPAD detector, triggers an avalanche of charge carriers creating a current signal. The current is later detected and converted to a digital signal; typically within less than 100ps. Good SPAD detectors have low jitter, deadtimes and dark count rates (DCR) while maintaining high photon detection probabilities (PDP).

Jitter refers to the temporal delay between the time of photon arrival and signal detection. Timing jitter is quantified by measuring delay Δt between the emission of a LASER pulse (recorded by a Laser Clock), and the detection signal from the SPAD. Accumulating these events in a histogram it is possible to extract the instrument response function (IRF). The jitter is then reported as the Full Width at Half Maximum (FWHM) of this distribution.

After an avalanche is triggered, the SPAD must reduce the voltage across the diode to stop the current in a process called quenching. A period of voltage recharge immediately follows to restore the detector ability to detect incoming photons. The time period between the start of the avalanche and the end of the recharge is referred to as **dead-time** because the detector remains blind to subsequent incident photons.

Not every incident photon successfully triggers a macroscopic avalanche. **PDP** quantifies the percentage of incident photons that result in a measurable electrical pulse. This indicator is dependent on the photon's wavelength and the detector's overvoltage.

SPADs will occasionally register false counts even in complete darkness. **DCR** counts are mainly triggered by thermally generated charge carriers inside the depletion region that initiate an avalanche indistinguishable from a true photon detection. Tunneling effects and lattice defects are also causes of DCR.

0.2 Statistical Properties of Light

Phase-coherent sources are defined by an electromagnetic field that oscillates with a precise, constant phase over time. A popular quantum model for these systems makes use of the second quantization formalism of the harmonic oscillator, resulting in a coherent state $|\alpha\rangle$. This state is the right eigenstate of the photon annihilation operator $\hat{a}$, such that $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$. The eigenvalue α is a complex number where its squared amplitude encodes the average number of photons ($|\alpha|^2 = \mu$), and its angle represents the macroscopic phase of the field.

To calculate the probability of detecting n photons, a projection of the coherent state

onto the photon number ¹ basis results in

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (1)$$

The probability $P(n) = |\langle n|\alpha\rangle|^2$ simplifies exactly to a Poissonian distribution. Because the phase is stable, individual photons behave completely independently; the arrival of one photon provides no predictive information about the next. Consequently, the variance equals the mean ($\sigma^2 = \mu$).

Non-coherent sources—such as stars, flames, or continuous lasers passed through rotating diffusers—exhibit super-Poissonian statistics. In this regime, the total electric field at the detector is the superposition of countless independent waves with random, fluctuating phases:

$$E(t) = \sum_i E_i e^{-i(\omega t + \phi_i(t))} \quad (2)$$

Because $E(t)$ is the sum of many random variables, the Central Limit Theorem dictates that the electric field becomes a Gaussian random variable. The absolute square of a Gaussian random field yields a macroscopic intensity $I(t)$ that strictly follows an exponential probability distribution:

$$P(I) = \frac{1}{\langle I \rangle} e^{-\frac{I}{\langle I \rangle}}. \quad (3)$$

The bosonic nature of photons in these chaotic fields manifests in a phenomenon called photon bunching. Because photons prefer to occupy the same modes, they arrive in clumps followed by dark lulls. This broadens the photon number distribution into a Bose-Einstein distribution, significantly increasing the statistical noise such that $\sigma^2 = \mu + \mu^2$.

Single-photon sources belong to a strictly quantum regime that cannot be described by classical waves. Because the source physically emits photons one at a time, the probability of detecting two photons simultaneously is zero. This forced temporal separation leads to photon anti-bunching, producing

¹Also called Fock basis.

a statistical distribution that is tighter than a coherent laser, characterized by $\sigma^2 < \mu$.

0.2.1 Second Order Coherence $g^{(2)}$

The second order correlation function $g^{(2)}$ quantifies intensity correlations, photon statistics and time separated probability of detecting two photons. For reference, a value for $g^{(2)}(0) < 1$, is an indicator of antibunching; a quantum characteristic of the light source. In contrast, an ideal coherent state yields $g^{(2)}(0) = 1$. While thermal light gives $g^{(2)}(0) = 2$.

To summarize, second order coherence serves as a tool that distinguishes the nature of a variety of light sources. This work aims to present the reader a experimental procedure that measures $g^{(2)}$, leveraging the power of SPAD and time tagger technology.

Equipment and Materials

Facilities

Room	Description	Notes
MC A3 153	Optics Lab	CW
MC A3 193	Rad/Optics Lab	Pulsed Laser

Lasers and Light Sources

- Pulsed LASER: PicoQuant PDL Series
 - **Optical Head:** PicoQuant Laser Diode Head
 - **Wavelength:** 405 nm
 - **Repetition Rate (f_{rep}):** Tunable discrete frequencies (operated between 1 MHz and 40 MHz).
 - **Emission Profile:** Picosecond pulsed regime.
- Pulsed LASER: NKT Photonics
 - Model: [Origami XP / Origami HP] (Femtosecond)
 - Wavelength: [] nm (e.g., 1030 nm or 515 nm)
 - Pulse Duration: [] fs
 - Repetition Rate: 50 MHz

- Continuous Wave (CW) LASER

- **Brand / Model:** Thorlabs S1FC660 (Fiber-coupled)
- **Wavelength:** 660 nm
- **Operating Power:** Analog tunable from 0 mW up to a maximum output of 19.8 mW.

- Pseudothermal Light Simulator

- Components: Rotating ground glass diffuser and pinhole aperture.
- Diffuser Motor Speed / Coherence Time: [] ms

Timing Electronics

- Swabian Instruments Time Tagger

- Model: Ultra, USB 3.0
- Max Data Transfer Rate: 90 MTags/s
- Max Input Frequency: 475 MHz
- Dead Time: 2.1 ns
- Base Jitter: 42 ps RMS (100 ps FWHM)
- Physical Inputs: 4 (50 Ω impedance)
- Trigger Level Range: -2.5 V to +2.5 V

Detectors and Optical Components

- SPAD Module

- Model: NV02TPX
- 64 bins of up to 5ns width

- Neutral Density (ND) Filters

- Optical Density (OD) values applied: []

- Connectivity

- SMA Cables (50 Ω impedance)
- SM1 FC2 (Fiber to Free Space Coupling)

Layout and Hardware Calibration

The experimental layout used through the measurements is presented figure 1. The layout is designed to allow interchanging light sources incident on the SPAD. Optical elements can also be introduced between the source and the SPAD for attenuation, modulation and alignment purposes.

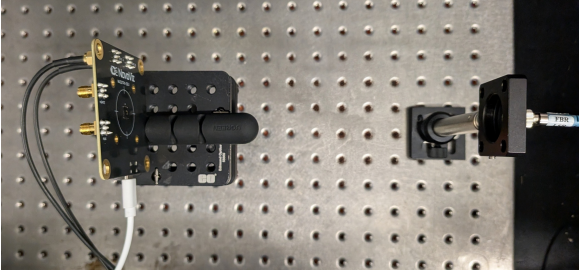


Figure 1: Experimental layout. Source is feed into the system with fiber optic cable that couples into free space with the SM1-FC2 and irradiates the SPAD.

0.3 SPAD Characterization

With the set up complete, a SPADs calibration is performed to set the operational baseline for the subsequent measurements. The SPAD is configured with a $V_{Op} = needtomeasure$ with the adjustment trimmer potentiometer and deadtime potentiometer. We fix the configuration mid-range for optimal performance and proceed to extract the performance indicators.

First, a dark count experiment where we run the `tt.CountRate` function. We integrate 205 measurements give dark counts of 9.74 cps and 10,301.49 cps for SPAD A and B respectively at this specific configuration. Next, we extract the dead time of the detector, and afterpulsing probabilities with the procedure listed below:

1. Connect the CW LASER source to the fiber optics line feeding the system.
2. Connect the SPAD detectors with SMA Cables to Swabian Time Tagger.

3. Initialize the autocorrelation measurement (`tt.Correlation`) using a single SPAD channel as both the "start" and "stop" trigger.
4. Wait for detection events to accumulate for 10 seconds.
5. Extract dead time and afterpulsing probabilities from data (See figure 2).
6. Repeat for 5 time to gather Statistics.

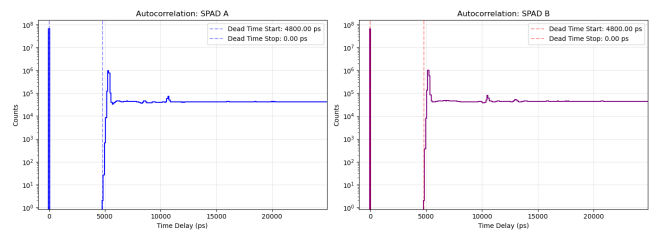


Figure 2: Autocorrelation histogram of the SPAD channels. Dead time is extracted from the temporal delay required for counts to recover from zero. Afterpulsing is extracted from taking the ratio of excess detections following the dead time to the total number of detections

The `tt.Correlation` time tags a series of detection events, calculates the time difference between all events (τ) and builds a histogram out of the number of detections that correspond to a particular time delay. Thus, from figure 2, we identify the total number of detections in bin $\tau = 0$ and the deadtime as the zero detection plateau. Afterpulsing is extracted as a ratio of excess detections that follow immediately after the deadtime to the total number of detections. The results are summarized in the table below:

Table 1: Summary of Detector Calibration Parameters

Parameter	SPAD 1	SPAD 2
DCR (cps)	9.74cps	10,301.49
Mean Dead Time (ns)	4.90 ± 0.12	4.85 ± 0.09
Max Theoretical Rate (MHz)	204.08	206.19
Afterpulsing Probability (%)	2.41	2.30

Poissonian Statistics Verification

This section presents the experiment and measurements of the Poissonian and Super-Poissonian probability distributions. We followed the procedure below:

- Feed the CW Laser into the optical setup in figure 1.
- Mount Filter in between the source and the SPAD.
- Define the temporal length of the windows integrating event counts: $100 \mu s$
- Connect SPAD to Swabian Time Tagger.
- Run the SWABIAN `tt.Counter` function and plot histogram [Swabian Instruments, 2026].

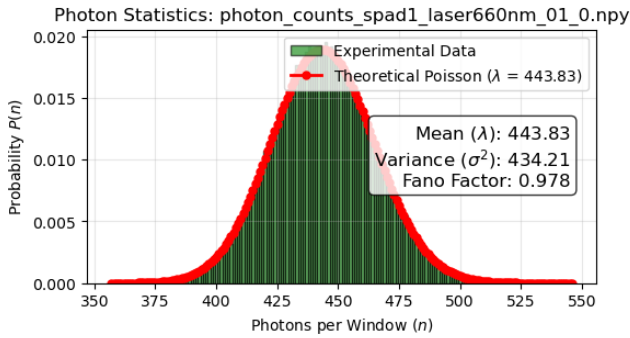


Figure 3: Photon counting statistics of the CW laser at $1mW$.

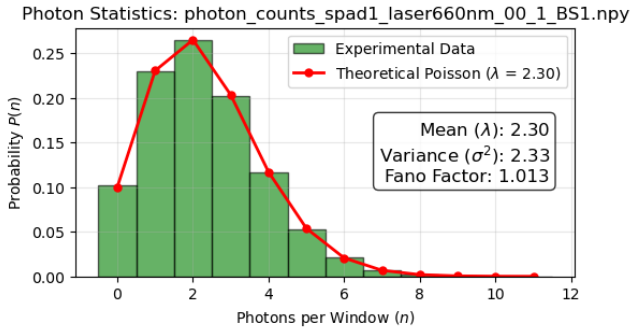


Figure 4: Photon counting statistics of the attenuated CW laser.

As shown in figure 4, the coherent laser source yields a poissonian distribution even as we vary the power of the CW Laser.

Furthermore, in figure 3, a polarizing beam splitter was introduced to measure the curve in low power regime; also resulting in a poissonian distribution. The fact that the variance matches the mean ($\sigma^2 \approx \mu$),², verifies the source as a coherent emitter.

0.4 Non Coherent Light

We proceed to switch the light source for a fluorescent light bulb and run the previous experiment with the CW Laser turned off. The resulting super-Poissonian distribution is shown in figure 5.

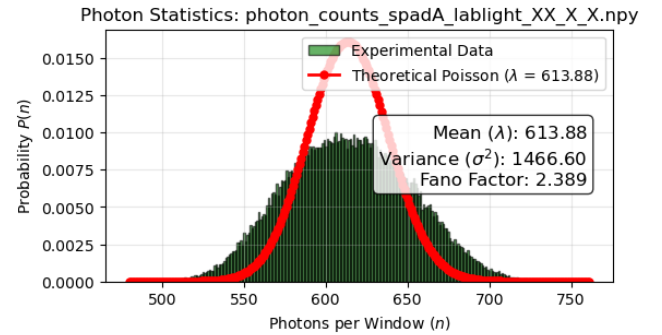


Figure 5: Photon counting statistics of the fluorescent lightbulb. The experimental data follows a super-Poisson distribution.

²Fano factor defined by $F = \frac{\lambda}{\sigma^2} \approx 1.0$.

Coincidence Measurements

0.5 Correlation Value

First measurements measure second order coherence at $\tau = 0$ i.e $g^{(2)}(0)$. The procedure is given bellow.

1. Feed the Pulsed Laser into the optical setup in figure 1.
2. Select a Laser frequency: $1/T_{dead} \approx 2 \cdot 10^8 = 200MHz$.
3. Select Exposure Time.
4. Connect Laser Sync and SPAD channels into the time tagger.
5. Measure timestamps of detection events.
6. Compare Channel Ticks.

$$\Delta t_{coincidence} = \Delta t_B - \Delta t_A$$

Coincidence if $\Delta t_{coincidence} \leq 2 \cdot Jitter$

7. Calculate Probabilities

$$P_{A|B} = \frac{\#Events_{A|B}}{\#Pulses}$$

$$P_{AB} = \frac{\#Coincidences}{\#Pulses}$$

8. Calculate Correlation

$$g^{(2)}(0) = \frac{P_{AB}}{P_A \cdot P_B}$$

9. Check 3, 4 fold correlations $g^{(3)}, g^{(4)}, \dots$

0.5.1 Pulsed Laser Measurements (05.06.2026)

Measurement of $g^2(\tau = 0)$ where taken with 10 ps resolution with a laser attenuation adjusted to obtain 2% of total pulses detected by SPADs. Results are summarized in Figure 6.

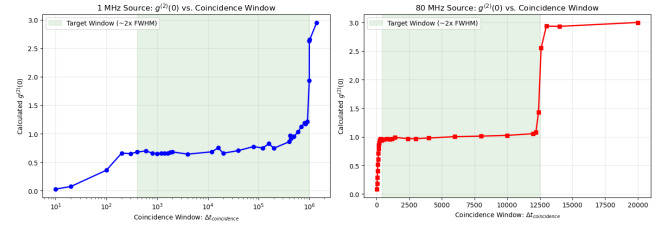


Figure 6: $g^2(0)$ measurement detailed in experiment 0.5.

These results were obtained in the laboratory session following the detailed steps bellow:

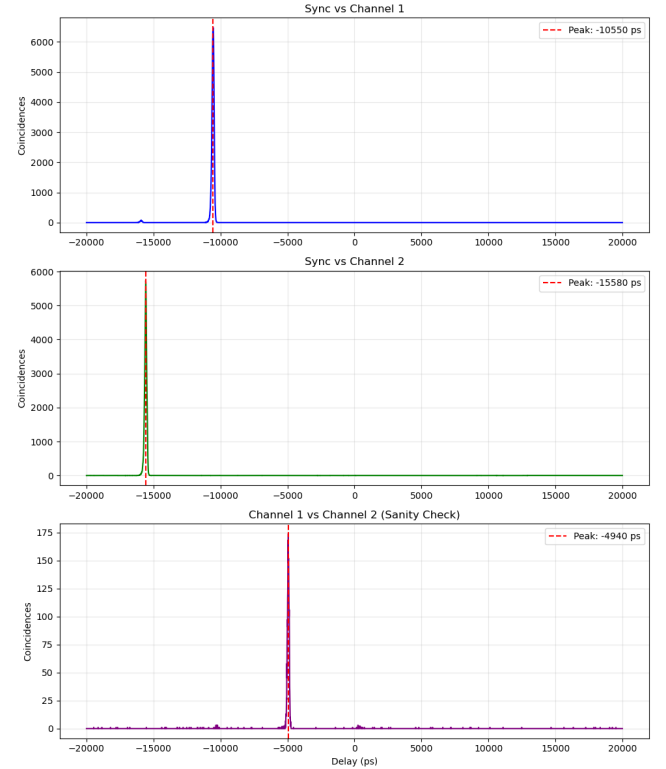


Figure 7: Differences in signal times. Calibration using `tt.Correlation` function

- Connect SPAD A, SPAD B, and Laser Sync to Swabian channel 1,2 and 3 respectively.

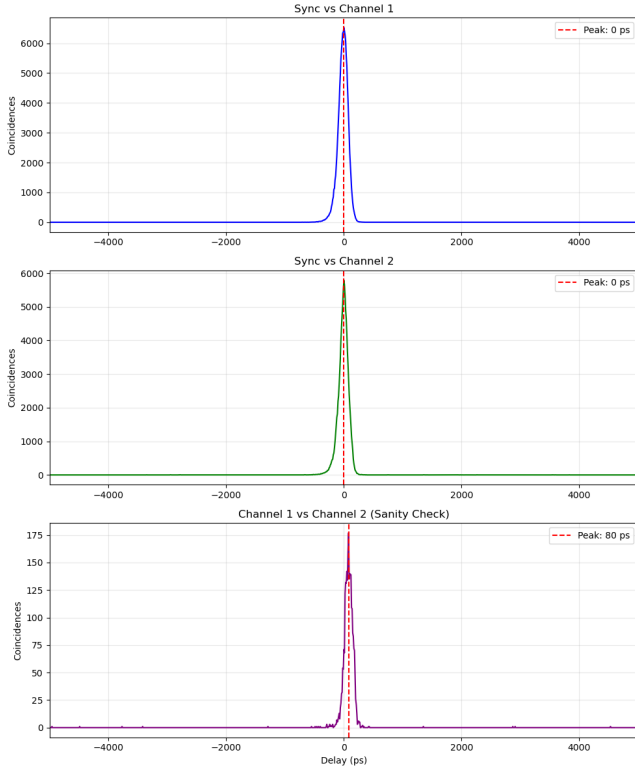


Figure 8: Calibrated signals to zero ps delay.

- LASER frequency is configured at 1MHz . Select internal oscillator *Source: Int2*, and Rep Frequency at *1*.
- Read the Pico-Quant LASER at 36.8% power.
- Measure Power at $3\mu\text{W}$
- Align the LASER to the SPADs active areas.
- Place ND0.6 Filter.
- Isolate the system from external light sources.
- Calibrate. Set baseline for SPAD dark counts and Laser clock. Laser head remains off.
 - Use `tt.Counttrate` to extract average counts-per-second (cps) over $5 * 10^{11}$ picoseconds.
 - Averaging the previous averages 65 times, we get:
 - * Mean SPAD A: 9.908 cps
 - * Mean SPAD B: 9,816.862 cps

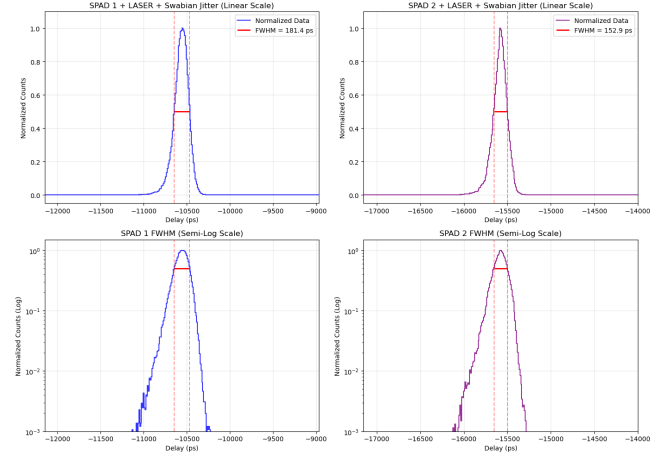


Figure 9: Optical Setup Jitter made up of Swabian, LASER & SPAD.

* Mean LASER Sync:
999,995.969 cps

- Turn Laser On and calibrate cps.
 - Use `tt.Counttrate` to extract average counts-per-second (cps) over $5 * 10^{11}$ picoseconds.
 - Averaging the previous averages 94 times, we get:
 - * Mean SPAD A: 26,262.771 cps (2.6% of pulses detected)
 - * Mean SPAD B: 30,842.627 cps (3% of pulses detected)
 - * Mean LASER: 999,995.819 cps
- Then, we proceed with the calibration for the optical path differences. We use `tt.Correlation` function, with 10ps resolution to find the time differences between two channels and obtained the following results.
 - Sync → SPAD A Delays:
[−10550] ps
 - Sync → SPAD B Delays:
[−15580] ps
 - SPAD A → SPAD B Delay:
[−4940] ps
 - Refer to Figure 7
- Input delays into Swabian `tt.setInputDelay(SPAD, delay)`.

- Refer to Figure 8
- Proceed with $g^{(2)}(0)$ measurement (Experiment procedure in subsection 0.5).
- Reference Results in Figure 6, left graph.

Aumenting the LASER Frequency

- Read the Pico-Quant LASER at 36.8% power.
- LASER frequency is configured at 80MHz. Select internal oscillator *Source: Int1*, and Rep Frequency at 1
- Measure Power at 100μW
- Turn Laser On and calibrate cps.
 - Use `tt.Counttrate` to extract average counts-per-second (cps) over $5 * 10^{11}$ picoseconds.
 - Averaging the previus averages 31 times, we get:
 - * Mean SPAD A: 839,914.581 cps (1%)
 - * Mean SPAD B: 683,993.507 cps
 - * Mean LASER: 79,999,739.641cps
- Then, we proceed with the calibration for the optical path differences. We use `tt.Correlation` function, with 10ps resolution to find the time differences between two channels and obtained the following results.
 - Sync → SPAD A Delays: [1970] ps
 - Sync → SPAD B Delays: [-3050] ps
 - SPAD A → SPAD B Delay: [-4930] ps
 - See *optical_path_calibration.png*
- Proceed with $g^{(2)}(0)$ measurement (Experiment procedure in subsection 0.5).
- Reference Results in Figure 6, right graph.

0.5.2 Pulsed Laser Measurement (05.15.2026)

In Figure 6, we did not obtain the expected result $g^{(2)}(0)$ we where expexitng. Probably this was die to the low operation power 3μW. To test this hypotesis, we make a new measurement at 1MHz with 33.5μW. Results are shown in Figure 10.

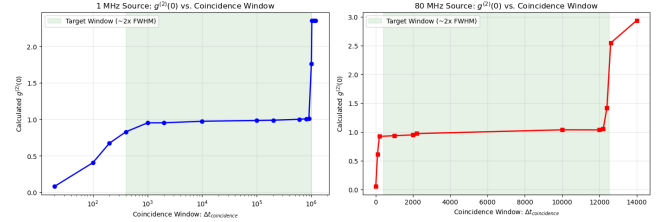


Figure 10: $g^{(2)}(0)$ measurement detailed in experiment 0.5.

- Connect SPAD A, SPAD B, and Laser Sync to Swabian channel 1,2 and 3 respectively.
- Read the Pico-Quant LASER at 100% power.
- LASER frequency is configured at 1MHz. Select internal oscillator *Source: Int2*, and Rep Frequency at 1
- Measure Power at 33.63μW.
- Align the LASER to the SPADs active areas.
- Isolate the system from external light sources.
- Calibrate. Set baseline for SPAD dark counts and Laser clock. Laser head remains off.
 - Use `tt.Counttrate` to extract average counts-per-second (cps) over $5 * 10^{11}$ picoseconds.
 - Averaging the previus averages 139 times, we get:
 - * Mean SPAD A: (Run 2: 9.252 cps, Run 3: 9.714 cps)
 - * Mean SPAD B: (Run 2: 9,542.935 cps, Run 3: 9,934.061 cps)

- * Mean LASER Sync: (Run 2: 999,996.000 cps, Run 3: 999,995.837 cps)
- Turn Laser On and calibrate cps.
 - Use `tt.CountRate` to extract average counts-per-second (cps) over $5 * 10^{11}$ picoseconds.
 - Averaging the previous averages 131 times, we get:
 - * Mean SPAD A: (Run 2: 665,354 cps, Run 3: 670,303.900 cps)(67% of pulses detected)
 - * Mean SPAD B: (Run 2: 584,594 cps, Run 3: 592,776.250 cps)(59% of pulses detected)
 - * Mean LASER: 999,996.000 cps
- Then, we proceed with the calibration for the optical path differences. We use `tt.Correlation` function, with 10ps resolution to find the time differences between two channels and obtained the following results.
 - Sync $\rightarrow$ SPAD A Delays: [−10140, −10190] ps
 - Sync $\rightarrow$ SPAD B Delays: [−15160, −15200] ps
 - SPAD A $\rightarrow$ SPAD B Delay: [−4910, −4930] ps
- Input delays into Swabian `tt.setInputDelay(SPAD, delay)`.
- We observe $g^{(2)}(0)$ value approaches 1 as we would expect. Figure ??

Aumenting the LASER Frequency

- Read the Pico-Quant LASER at 36.8% power.
- LASER frequency is configured at 80MHz. Select internal oscillator *Source: Int1*, and Rep Frequency at 1
- Measure Power at 100μW

- Turn Laser On and calibrate cps with ND0.6 filter until we reach the 1% domain.
- We continue the calibration for the optical path differences using the `tt.Correlation` function, with 10ps resolution.
 - Sync $\rightarrow$ SPAD A Delays: [1970] ps
 - Sync $\rightarrow$ SPAD B Delays: [−3050] ps
 - SPAD A $\rightarrow$ SPAD B Delay: [−4930] ps
- Proceed with $g^{(2)}(0)$ measurement and obtained the results shown in Figure ??.

As a secondary result, we extract the system jitter from the Figures 7 and 8. As shown in Figure 9, jitter values for SPAD A and B are 181.4&152.1 respectively.

0.6 Correlation Function

Second measurements designed to extract the second order coherence function $g^{(2)}(\tau)$.

1. Feed the Pulsed Laser into the optical setup in figure 1.
2. Select a Laser frequency: $1/T_{dead} \approx 2 \cdot 10^8 = 200MHz$.
3. Build Histogram of $G^{(2)}(\tau)$
4. Normalize histogram by the average count of the time bins far from $\tau = 0$ to get $g^{(2)}(\tau)$.

We expect $g^{(2)}(0) = 1$ with a comb structure for the Pulsed Laser. In contrast, for thermal sources, $1 < g^{(2)}(0) \leq 2$ and decays over coherence times. The width of this decay is a measure of the coherence time ³.

0.6.1 Pulsed Laser Measurements (05.06.2026)

Results are summarized in Figure 11.

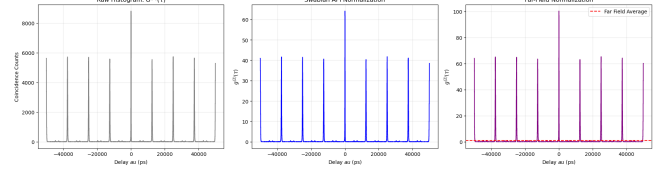


Figure 11: Autocorrelation function measurement.

A similar result can be obtained for the $1MHz$ signal with its characteristic period shown in Figure 12.

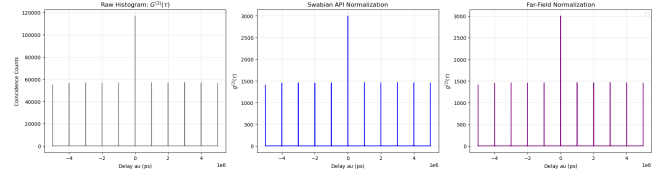


Figure 12: Autocorrelation function measurement.

However, both figures fail to quantify the normalized $g^{(2)}$ because the normalization is taking into account the secondary peaks. However, when we adjust the temporal window of the correlation analysis, we recover the results expected as discussed in 0.5. Refer to Figures 13 and 14.

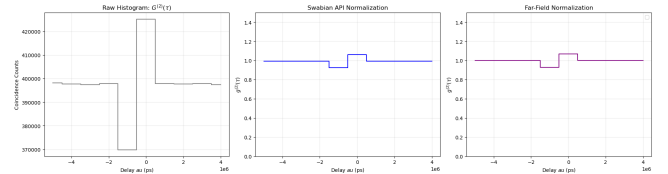


Figure 13: Autocorrelation function measurement.

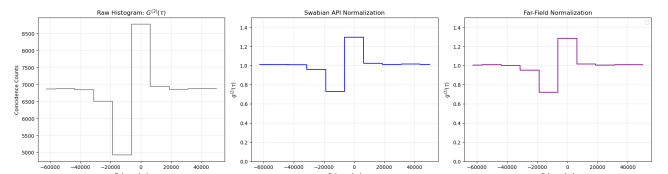


Figure 14: Autocorrelation function measurement.

³Set by the diffuser speed

Next we turn the attention to the thermal sources. We shine a thermal lamp into the SPAD and obtain its correlation function.

- Place 2.3 ND filter over the SPAD
- Turn on the Thermal Lamp or the Laboratory Light.
- Measure cps: 1,000,000
- Measure $G^{(2)}$, $g^{(2)}$
- Repeat for increasing cps.

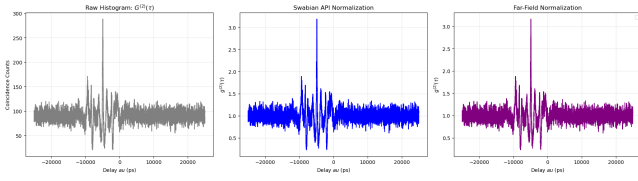


Figure 15: Autocorrelation function measurement.

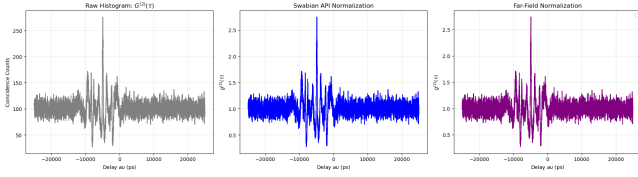


Figure 16: Autocorrelation function measurement.

We observe that lower cps, increase the signal to noise ratio.

Conclusion

This project established an experimental framework to characterize Single Photon Avalanche Diodes (SPADs) and classify the statistical properties of various light sources. Hardware baselining accurately quantified intrinsic detector parameters, including dark count rate, dead time, afterpulsing, and jitter. The setup successfully validated expected Poissonian distributions for coherent lasers and super-Poissonian distributions for thermal sources. Time-Correlated Single Photon Counting (TCSPC) enabled precise second-order coherence ($g^{(2)}$) measurements to further differentiate emission regimes. Careful optical path calibration ensured high-resolution detection of true coincidence events. Ultimately, this methodology accurately captured intrinsic source statistics while accounting for detector artifacts, clearly distinguishing between classical and non-classical light.

References

- [Antolovic et al., 2018] Antolovic, I. M., Bruschini, C., and Charbon, E. (2018). Dynamic range extension for photon counting arrays. *Optics Express*, 26(17):22234–22248.
- [Charbon, 2014] Charbon, E. (2014). Single-photon imaging in complementary metal-oxide semiconductor processes. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 372(2012):20130100.
- [Hofbauer et al., 2018] Hofbauer, M., Steindl, B., and Zimmermann, H. (2018). Temperature dependence of dark count rate and afterpulsing of a single-photon avalanche diode with an integrated active quenching circuit in 0.35 μm CMOS. *Journal of Sensors*, 2018.
- [Swabian Instruments, 2026] Swabian Instruments (2026). Measurement classes: Correlation — time tagger user manual.